Blue Earth Valley Power, Grid Modernization

Re-battery 16 controls before December and reset the receiving reclosers before June

To: Shawn Hudson, Debra Reed. From: Nathan Reeves. 11 November 2026

Bottom line. Replace the batteries in 16 recloser controls before December. Before June, move the receiving reclosers on the transfer directions the ADMS does not switch to settings that carry a transfer, rather than waiting for the four reinforcements. Run as they stand, the loops should avoid about 24,679 sustained customer interruptions in 2027 (80 per cent range 17,912 to 31,580); from January to October 2026 they avoided 21,328 on the order's definition.

1. Before this winter: batteries

A control on a dead line trips and closes on its battery, drawing 30 A, and stops below 21.2 V. The weekly load tests run at 5 A, and the annual functional tests run on line power, so neither shows how a weak battery behaves at -27 C. Fitting each battery's resistance to the temperature of its test hour and its age, and carrying it to -27 C at the end of February, the following would not operate: RCL-20149, RCL-20188, RCL-20205, RCL-20275, RCL-20355, RCL-20466, RCL-20536, RCL-20542, RCL-20547, RCL-20732, RCL-20977, RCL-21316, RCL-21357, RCL-21377, RCL-21643, RCL-21710.

2. Before next summer: transfer settings

On the directions whose receiving controls the ADMS does not move to group 2 (CTL-4C controls), the combined load can trip the receiving recloser after a transfer, with no fault found; operators switching the same transfers set group 2 first. Replaying the 2025 and 2026 summer loads at the hours the loops' lockouts strike, transfers in June to September 2027 would interrupt about 388 of the receiving feeders' own customers if the settings stand. The four reconductoring projects would remove part of that, but the capital works record puts comparable projects a median 3.9 months late. Directions to change: EGL212 to EGL292, EGL292 to EGL212, GTH587 to RPD436, RPD436 to GTH587, KST234 to MNP352, MNP352 to KST234, KST254 to LMV593, LMV593 to KST254, KST259 to LMV562, LMV562 to KST259, KST273 to LMV582, LMV582 to KST273, LMV580 to KST217, KST217 to LMV580, MDL330 to EGL259, EGL259 to MDL330.

3. 2026 on the commission's definition

2026 (Jan to Oct)	Customer interruptions
Restored within five minutes of the first loss of supply	25,056
Restorations undone by a receiving recloser trip	-2,407
Receiving feeders' own customers interrupted	-1,321
Avoided, order para. 2	21,328

Sources: SCADA device events, OMS, ADMS FLISR archive; customers from the outage record's steps.

The event archive drops events when its queue overflows in storms, and the front end ran so slowly then that none of those restorations came inside five minutes; the device record shows what the scheme did. The vendor dashboard counts completed sequences in the archive whatever their duration.

4. 2027 as the loops stand

At the record's average rate of lockouts since January 2022, with the scheme's own success rate, the batteries and the transfer trips as they stand and the reinforcements at their likely dates: about 24,679 customer interruptions avoided, 80 per cent range 17,912 to 31,580. The workbook carries every figure and its working.